graded_bets_max_legacy

equity strategy | family: equity-max | status: reference | derived from `graded_bets_max`

What this strategy does

graded_bets with bigger bets for more return: base unit 0.005, 6x leverage cap -- ORIGINAL leverage cap (new positions only, open book ignored)

Why it should work. Same drift as Extreme Picks but harvested across the whole cross-section for lower variance.

Design notes (from the strategy's source file).

Max-size variants ("<name>_max") of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.005
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	proportional
exposure_cap	entry_cohort
beta_overlay	0.0

Performance

metric	graded bets max legacy	market (CRSP VW)
Annualized return	10.84%	9.94%
Annualized volatility	12.40%	19.35%
Sharpe (NAV-pct)	0.89	0.59
Sortino	1.28	
Max drawdown	-38.8%	-55.5%
Calmar	0.28	
Final NAV ($10M start)	$74,180,822	
Worst day	-8.03%	
% positive days	51.5%	

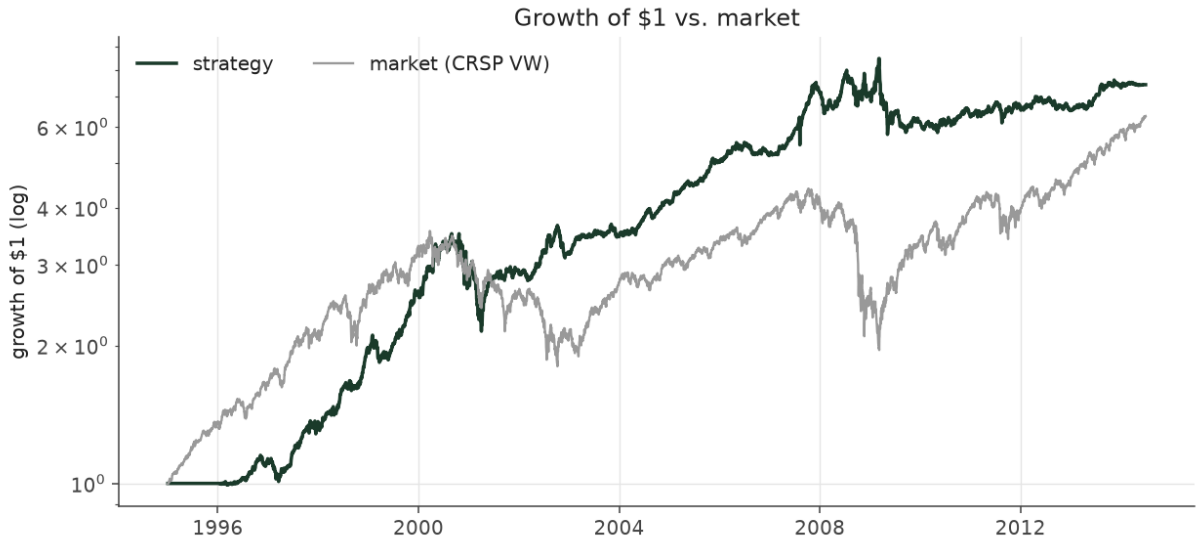
Sharpe 1996-2001	1.24	
Sharpe 2002-2007	1.63	
Sharpe 2008-2013	0.09	

Correlation of daily returns with the market: **0.03**.

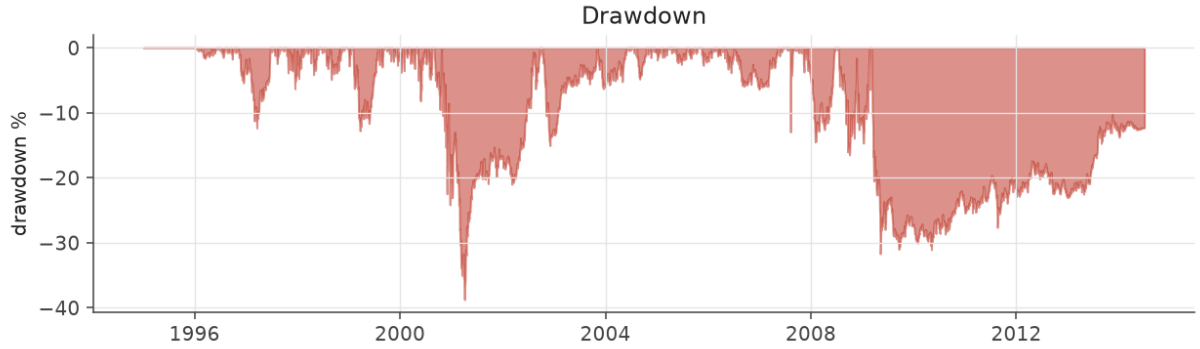
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 0.892

• best return among equity: `market_boost_max` (0.116); this one: 0.108

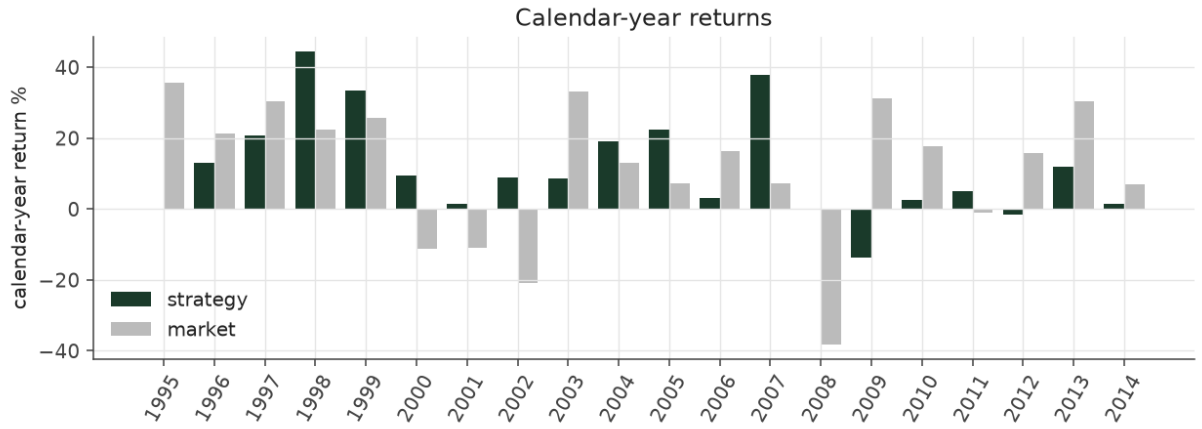
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.28



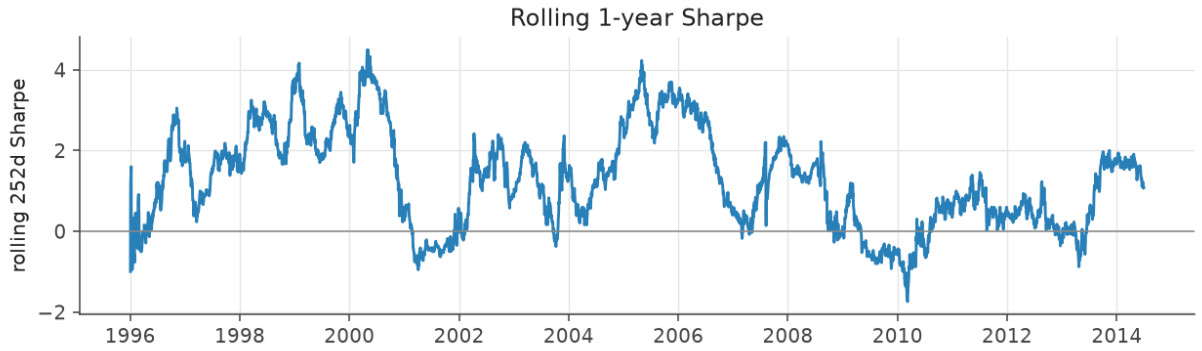
NAV (log scale)



Drawdown from running peak

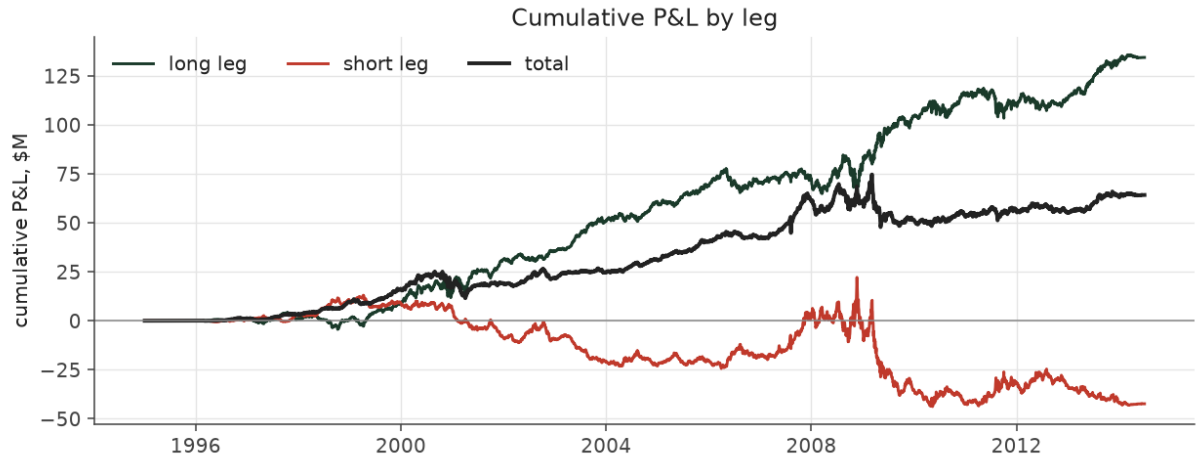


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	134.38	-42.37
P&L after costs ($M)	120.38	-56.20
of which common (average-event) component ($M)	124.23	-110.08

of which excess / PEAD component ($M)	9.69	68.09
excess minus costs ($M)	-4.29	54.28
gross return per $ traded (bp)	161.80	-56.51
positions in the money (hit rate)	0.55	0.48
stand-alone Sharpe (leg only)	0.94	-0.18

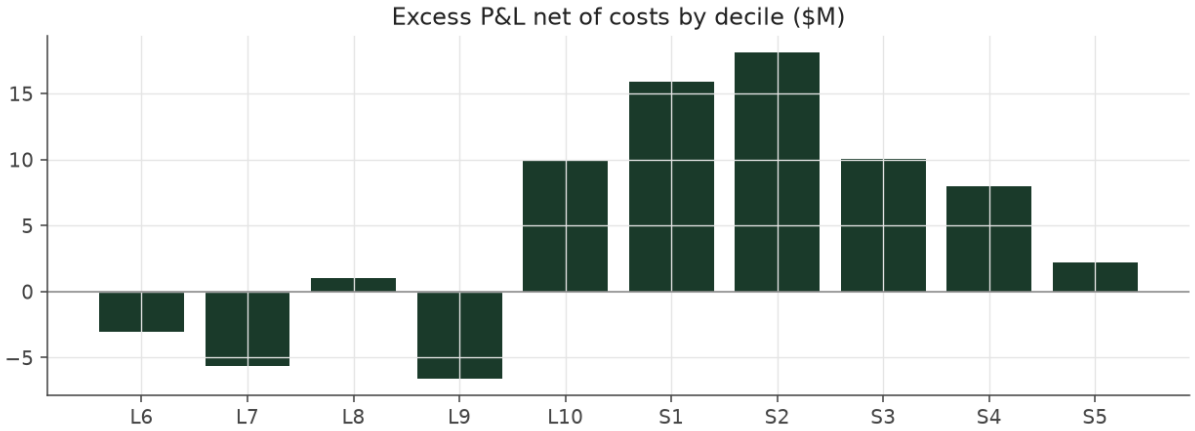
Daily P&L correlation between the two legs: **-0.77**.

• **The short leg loses money in absolute terms** (-42.4M before costs) while the long leg earns +134.4M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 124.2M / 110.1M. Stripping that common component, the surprise signal itself earned +9.7M on the long side and +68.1M on the short side.

• The two legs' daily P&L are -0.77 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.94 long / -0.18 short vs 0.89 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	6	18,256	446.78	2.58	-2.16	0.87	-3.03	-6.1%	-67.91
long	7	17,431	1,034.12	12.39	-3.81	1.85	-5.66	-11.3%	-54.75
long	8	16,588	1,679.35	27.38	3.90	2.86	1.04	2.1%	6.20
long	9	17,747	2,343.83	26.13	-2.79	3.83	-6.62	-13.2%	-28.24
long	10	17,174	2,801.07	51.89	14.55	4.58	9.97	19.9%	35.60
short	1	17,169	2,357.06	-16.78	20.22	4.32	15.90	31.8%	67.45
short	2	17,281	2,060.53	-13.14	21.82	3.71	18.11	36.2%	87.90
short	3	15,970	1,559.44	-13.10	12.92	2.89	10.03	20.1%	64.30
short	4	21,469	1,213.21	-9.67	10.29	2.29	8.00	16.0%	65.96
short	5	14,319	307.57	-3.50	2.83	0.60	2.23	4.5%	72.34



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $

1	19,578	681.66	3.56	8.34	2.41	5.93	11.9%	86.94
2	32,212	2,419.67	14.98	22.47	6.28	16.18	32.4%	66.89
3	36,915	3,811.99	20.93	25.43	7.56	17.87	35.8%	46.88
4	40,327	4,383.17	12.96	12.87	6.64	6.22	12.5%	14.20
5	44,372	4,506.47	11.76	8.69	4.93	3.76	7.5%	8.34

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	19,171	404.49	5.73	8.48	2.02	6.46	12.9%	159.65
2	32,362	2,234.14	12.87	22.01	6.70	15.31	30.6%	68.51
3	37,415	4,138.94	21.90	30.16	8.28	21.88	43.8%	52.86
4	40,575	4,479.25	10.63	10.88	6.27	4.61	9.2%	10.29
5	43,876	4,545.40	13.07	6.24	4.54	1.70	3.4%	3.73

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	77,748	6,570.82	-7.15	13.24	10.82	2.42	4.8%	3.67
41-60d	33,200	3,470.48	17.71	14.64	6.18	8.46	16.9%	24.39
61-90d	48,100	4,513.69	18.35	21.08	8.24	12.84	25.7%	28.45
>90d	14,356	1,247.97	35.26	28.82	2.58	26.24	52.5%	210.29

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	50,364	2,077.53	18.41	18.87	3.65	15.21	30.4%	73.23
2002-07	58,164	5,345.18	38.34	46.08	9.31	36.77	73.6%	68.80
2008-13	64,876	8,380.25	7.43	12.83	14.86	-2.03	-4.1%	-2.42

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	32,133	2,767.05	15.00	6.05	4.85	1.21	2.4%	4.36
Chems	3,926	372.28	1.82	2.61	0.62	1.99	4.0%	53.51
Durbl	4,066	360.55	1.02	2.28	0.65	1.63	3.3%	45.20
Enrgy	9,095	851.68	9.11	11.08	1.38	9.70	19.4%	113.91
Hlth	17,555	1,448.28	9.86	7.55	2.71	4.84	9.7%	33.41
Manuf	17,051	1,579.69	2.10	5.39	2.72	2.66	5.3%	16.86
Money	25,815	2,535.78	7.39	18.28	4.45	13.84	27.7%	54.56
NoDur	7,510	686.61	2.43	2.25	1.17	1.09	2.2%	15.84

Other	26,581	2,448.64	9.02	12.50	4.62	7.88	15.8%	32.19
Shops	17,607	1,587.20	7.84	7.85	2.77	5.08	10.2%	31.99
Telcm	5,624	491.57	-1.29	0.74	0.82	-0.08	-0.2%	-1.55
Utils	6,441	673.63	-0.12	1.20	1.07	0.12	0.2%	1.83

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	57,238	5,515.57	11.43	17.20	8.91	8.28	16.6%	15.02
low vol	58,337	5,073.50	25.96	31.12	9.67	21.45	42.9%	42.27
mid vol	55,592	5,058.35	26.71	29.59	8.92	20.67	41.3%	40.86

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	10.84%	0.00%
0.5%	7.5	10.23%	0.60%
1.0%	14.9	9.56%	1.27%
2.0%	29.9	7.94%	2.90%
5.0%	74.7	-100.00%	110.84%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.